John von Neumann, The Computer & the Brain, Third Edition (2012); the file contains the complete two-part structure: Part 1 - The Computer and Part 2 - The Brain, with sections on control, codes, precision, memory hierarchy, neurons, stimulation, memory, mixed analog/digital processing, and the statistical character of neural signaling. The edition history in the file gives first edition 1958, second edition 2000, and third edition 2012.

What I extracted for COS

The most important idea is not “make COS look like a brain.” It is:

> A cognitive machine should be designed around the relationship between representation, computation, control, memory, timing, reliability, and parallel organization.

1. COS needs both digital and analog computational domains

Von Neumann distinguishes digital and analog procedures and also discusses machines in which the two interact.

For COS this suggests a clean separation:

DISCRETE / SYMBOLIC DOMAIN
- instructions
- logic
- proofs
- states
- authorization
- exact mathematics

CONTINUOUS / APPROXIMATE DOMAIN
- sensors
- physical quantities
- probabilities
- numerical simulation
- neural-style activations
- uncertain measurements

↕

HYBRID INTERFACE

This fits very naturally with our existing exact mathematics + explicit approximate/error layer. The book strongly reinforces that these should not be confused.

2. Memory is not one thing

The computer section develops a memory hierarchy in which capacity and access time trade against each other. The point is architectural: different classes of memory exist because not every piece of information needs the fastest access.

For COS, memory should therefore become an explicit hierarchy:

L0 registers / immediate state
L1 working cognitive state
L2 active knowledge
L3 verified knowledge store
L4 long-term memory
L5 archival/source memory
L6 external world / devices

And each memory object should carry things such as:

address
size
access_cost
latency
persistence
integrity
provenance
priority

This is more than an optimization. Memory access belongs in the cognitive execution model.

3. Logical depth is a major COS concept

The brain discussion repeatedly contrasts logical depth with the ability to process many things simultaneously. The book's argument is that a system with relatively slow elementary components can compensate through enormous parallel breadth.

For COS, I would formalize:

DEPTH = number/cost of dependent sequential stages
BREADTH = amount of work that can proceed concurrently

Then the planner can reason about:

total cognitive cost
≈ sequential depth
+ parallel workload
+ communication
+ synchronization
+ memory access

That connects Book 4 directly with the Shape / Topology / Placement layer extracted from Book 3.

4. “Code” has two levels

One of the strongest ideas in the book is the distinction between a complete code and a short code. A short code makes one machine behave as though it were another machine by interpreting its higher-level instructions.

This is extremely important for COS.

It gives a conceptual foundation for:

Human intent
↓
COS semantic language
↓
virtual machine / interpreter
↓
COS execution model
↓
physical machine

So COS-B should remain only one language in a larger language stack.

Our current:

COS-B
↓
Bytecode
↓
CIR
↓
SIR

is therefore moving in exactly the right architectural direction. The next long-term step is to make virtual machine semantics explicit, so that a higher-level COS machine can be implemented on many different physical machines without losing meaning.

5. Memory may reside in changing structure

In the brain section, von Neumann discusses several hypotheses for how memory might be physically embodied, including changing stimulation thresholds, changing connections, genetic information, and active feedback structures. He explicitly presents these as uncertain possibilities.

The COS lesson is architectural rather than biological:

STATE OF THE SYSTEM
↓
can itself encode information

Therefore COS memory should not be restricted to a dictionary of stored facts.

There should eventually be:

symbolic memory
episodic memory
parameter/state memory
structural memory
connection/graph memory
procedural memory
source/provenance memory
capability memory

6. Reliability and precision are different resources

The book makes a particularly interesting observation about neural signaling: a statistical representation may have lower numerical precision while being more tolerant of individual message errors.

That gives us a useful COS design principle:

PRECISION ≠ RELIABILITY

COS should therefore track separately:

value
uncertainty
precision
confidence
integrity
verification status
error tolerance

Our v1.9 approximate/error system and v2.1 elementary-function domain/error handling should eventually grow into this more general framework.

7. The brain does not have to use human mathematical notation internally

Von Neumann argues that the brain's internal message system may be structurally different from ordinary human mathematics and language, with shallower logical/arithmetic depth and statistical forms of signaling.

For COS this suggests a very important rule:

Human language is an interface, not necessarily the native cognitive representation.

So:

English / Urdu / Hindi / Arabic / etc.
↓
Human Intent
↓
COS internal representation
↓
mathematical / logical / probabilistic / graph domains

This strongly supports keeping SIR and the underlying knowledge representations independent of natural-language wording.

8. Control is as important as computation

The first part spends substantial attention on logical control, memory-stored control, mixed control, and the way a machine's code determines its organized behavior. The contents themselves identify these as central sections of the computer half.

For COS:

COMPUTATION
is not enough

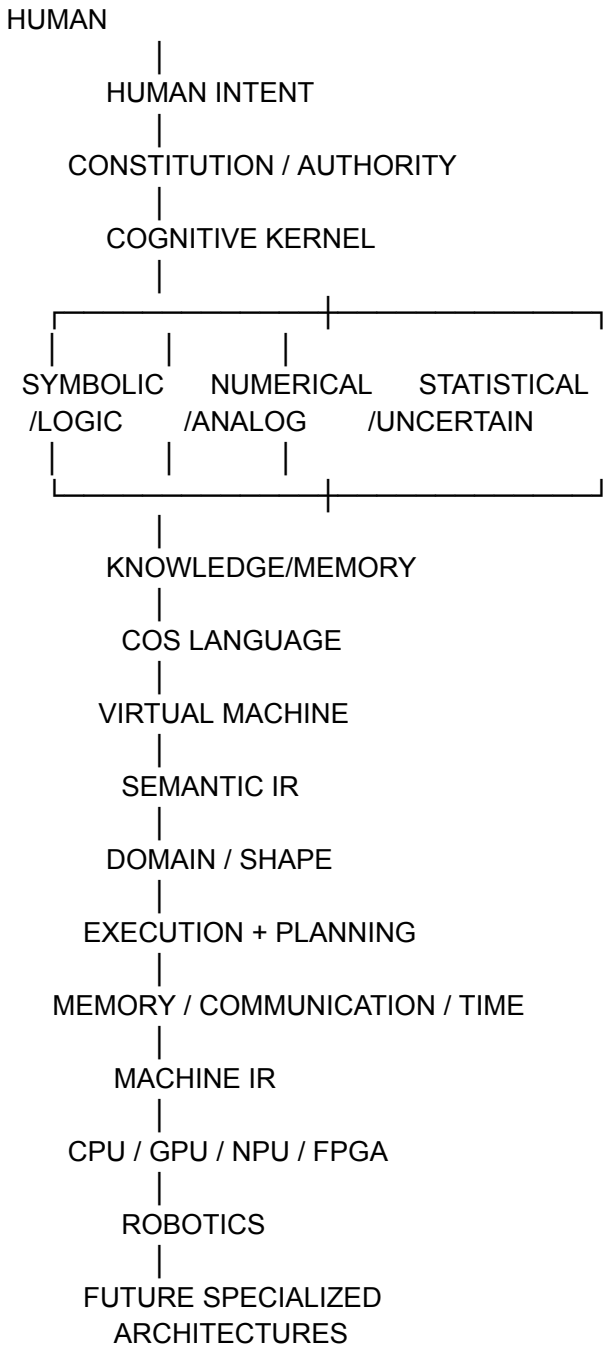
COMPUTATION
+
CONTROL
+
STATE
+
MEMORY
+
COMMUNICATION
=
AUTOMATON

This fits our constitutional/authorization layer especially well: what COS can compute is different from what COS is authorized to do.

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Book 4 contribution to the COS architecture

After Books 1-4, I would now define the conceptual stack as:




And surrounding the whole system:

PROVENANCE
VERIFICATION
INTEGRITY
OBSERVABILITY
ERROR MODEL
RESOURCE MODEL
RECOVERY

Four new COS architectural principles from Book 4

P24 - Computation may be hybrid.
COS must support discrete, continuous, and mixed computational domains.

P25 - Memory is hierarchical and cost-aware.
Memory location, latency, capacity, persistence, and integrity belong to execution planning.

P26 - Cognitive performance has depth and breadth.
COS should explicitly model sequential dependency depth and parallel computational breadth.

P27 - A machine can execute another machine's abstraction.
COS should support virtual-machine semantics and translation layers while preserving behavioral traceability.

One especially important conclusion

Books 1-3 taught us:

physical reality
→ abstraction
→ parallel organization

Book 4 adds:

representation
→ memory
→ control
→ hybrid computation
→ depth vs breadth
→ reliability
→ machine-within-machine

So COS is becoming less like a conventional “AI program” and more like what we originally intended:

a layered cognitive computing architecture that can express a task independently of the particular physical computer executing it.

I would not modify COS v2.1 yet. Book 4 gives us architectural laws to carry forward; implementation should wait until the next book is studied so we can avoid repeatedly redesigning the kernel.